

Finite denominator searches for an optimal degree-four rule

Research note, 30 September 2026

The bound in `optimized_two_adic_filter.tex` gives $R_4 \geq 16$, where repeated rational nodes are allowed. Its equality-case corollary shows that a putative 16-node rule has only odd denominators. If D is a common denominator, the second and fourth moments force $3 \mid D$ and $5 \mid D$. Thus D is an odd multiple of 15.

We exhaustively searched the grids $x_i \in \{0, 1/D, \dots, 1\}$ for $D = 15, 45, 75, 105$, finding no rule. This is a finite-grid result only; it does not improve the universal bound $R_4 \geq 16$.

Set $z_i = D(2x_i - 1)$. These are odd integers in $[-D, D]$, and the required integer equations are

$$\sum z_i = \sum z_i^3 = 0, \quad \sum z_i^2 = \frac{16D^2}{3}, \quad \sum z_i^4 = \frac{16D^4}{5}.$$

The program enumerates sorted multisets of sixteen positive magnitudes, split into their smaller and larger halves of eight. It matches their second and fourth power sums exactly. For each resulting multiset, a second meet-in-the-middle search enumerates all signs, matching first and third power sums. Repetitions and endpoints are included. The pruning inequalities use only the order of the two halves and lower or upper bounds for their remaining magnitudes.

D	stored smaller halves	visited larger halves	exact even-moment candidates
45	391619	3021351	2235
75	13189332	113553961	30120
105	149835139	1358156584	394429

Every candidate failed the exhaustive sign assignment check. The $D = 15$ grid is contained in each of these grids. The separate, less optimized program `search_degree4_magnitudes.cpp` independently reproduced all 2235 candidates at $D = 45$ and the same negative result.

The main implementation is `search_degree4_ordered_halves.cpp`; the machine-readable report is `degree4_grid_search.json`. No floating-point calculation enters either exact search.